

1 Question 2

- (a) Show that the *Independent Set Problem* in planar graphs can be solved in $\mathcal{O}(6^k n)$ time using branching.

We know that each node of a planar graph has a degree of at most 5. Therefore we can test at each step, each of the 6 possible combinations of nodes to include in the independent set. Assuming that a branch has a depth of k , then we have found an independent set of size at least k . Therefore, in the worst case, we have to test all possible combinations of nodes to include in the independent set, which is 6^k . Lastly, because the graph might not be connected, we have to repeat this process for each of the connected components in the graph, which is then in the worst case n times. Therefore, we have $\mathcal{O}(6^k n)$ time complexity.

2 Question 3

- (a) Show that the *Independent Set Problem* in planar graphs has a linear sized kernel.

Given that every planar graph is at 4-colorable, we know that if $n \geq 4 \cdot k$, then we can always find an independent set of size at least k . In that case, we can return yes immediately.

Otherwise, we have $n < 4k$. In that case, we leave the instance unchanged. Hence the remaining instance has at most $4k - 1$ vertices, which is linear in k .